

Title of the experiment

Y. O. Uname, partners: P. A. RtnerA, and P. A. RtnerB

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Abstract

In this experiment we studied a very important physical effect by measuring the dependence of a quantity V of the quantity X for two different sample temperatures. Our experimental measurements confirmed the quadratic dependence $V = kX^2$ predicted by Someone's first law. The value of the mystery parameter $k = 15.4 \pm 0.5$ s was extracted from the fit. We found that this value is 20% below theoretically predicted $k_{theory} = 17.34$ s. We attribute this discrepancy to low efficiency of our V -detector.

1 Theory overview

Here give a brief summary of the physical effect of interest and provide necessary equations. Here is how you insert an equation. According to references [?, ?, ?] the dependence of interest is given by

$$u(\lambda, T) = \frac{8\pi hc\lambda^{-5}}{e^{hc/\lambda kT} - 1}, \quad (1)$$

where T is temperature in Kelvin, c is the speed of light, etc. Don't forget to explain what each variable in the equation means, when you introduce it for the first time!

2 Experimental setup and procedures

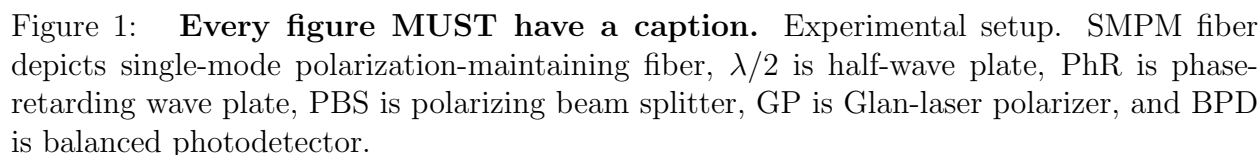
Note: LaTeX will put figures and tables at the locations where it thinks it is the best. Do not fight it, unless you really need it.

Give a schematic of the experimental setup(s) used in the experiment (see figure ??). Give the description of abbreviations either in the figure caption or in the text. Write a description of what is going on.

Don't forget to list all important steps in your experimental procedure!

Use active voice either in past or present through all the report and be consistent with it: The laser light comes from to ... and eventually arrived to the balanced photodiode as seen in the figure ??.

Sentences in the past voice while correct are generally considered hard to read in large numbers. The laser light was directed to ..., wave plates were set to ... etc.



In this section you will need to show your experimental results. Use tables and graphs when it is possible. Table ?? is an example.

Analysis of equation ?? shows ...

If in the process of data analysis you found any noticeable systematic error(s), you have to explain them in this section of the report.

2

Table 1: Every table needs a caption

Polarization	Target	Bin	$\langle x \rangle$	$\langle Q^2 \rangle$	$A_{\perp}^{meas}$	$\Delta A_{\perp}$
−	LiD	1	0.0233323	0.8429978	0.0044151	0.0030871
		2	0.0638046	1.5017358	0.0021633	0.0021343
		3	0.1892825	3.1877837	0.0006640	0.0022467
		4	0.4766562	7.1827556	-0.0197585	0.0085528
	NH ₃	1	0.0232572	0.8454089	0.0003600	0.0018642
		2	0.0633156	1.4870013	0.0023831	0.0013287
		3	0.1923955	3.1753302	-0.0024246	0.0013771
		4	0.4830315	7.3245904	-0.0284834	0.0047061
+	LiD	1	0.0233503	0.8340932	-0.0086018	0.0031121
		2	0.0638688	1.4785886	-0.0018465	0.0021452
		3	0.1892192	3.1277721	-0.0017860	0.0022525
		4	0.4778486	7.0313856	-0.0041773	0.0084659
	NH ₃	1	0.0232964	0.8439092	-0.0022961	0.0018851
		2	0.0633764	1.4814540	0.0021355	0.0013354
		3	0.1924094	3.1580557	-0.0065302	0.0013775
		4	0.4825868	7.3191291	-0.0290878	0.0047329

discussion. For example, it is easy to conclude that the experiment and theory match each other rather well if you look at Fig. ?? and Fig. ??.

4 Necessary remarks

Context of this template is out of sync with figures, tables, references, etc. It is just a template with examples how to use particular L^AT_EX features.

5 Conclusions

Here you briefly summarize your findings.

References

- [1] A. C. Melissinos and J. Napolitano, *Experiments in Modern Physics*, (Academic Press, New York, 2003).
- [2] N. Cyr, M. Têtu, and M. Breton, IEEE Trans. Instrum. Meas. **42**, 640 (1993).
- [3] *Expected value*, available at http://en.wikipedia.org/wiki/Expected_value.

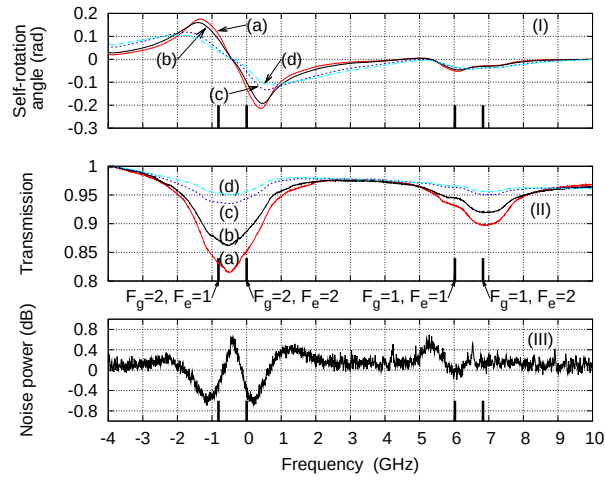


Figure 2: **Every figure MUST have a caption. Every plot MUST have axes labeled.** The dependence of self rotation and squeezing on the laser detunings.